

week_12\week12\src\main\java\week12\httpTttServer.java

```
1  package week12;
2  import com.google.gson.Gson;
3  import com.sun.net.httpserver.*;
4
5  import java.io.*;
6  import java.net.*;
7
8  public class httpTttServer {
9      public static void main(String[] args) throws IOException{
10         HttpServer server = HttpServer.create(new InetSocketAddress(9040), 0);
11         server.createContext("/move", new HttpHandler() {
12             @Override
13             public void handle(HttpExchange exchange) throws IOException {
14
15                 // CORS headers first
16                 exchange.getResponseHeaders().add("Access-Control-Allow-Origin", "*");
17                 exchange.getResponseHeaders().add("Access-Control-Allow-Methods",
18 "GET, POST, PUT, DELETE, OPTIONS");
19                 exchange.getResponseHeaders().add("Access-Control-Allow-Headers",
20 "Content-Type, Authorization");
21
22                 // Handle preflight
23                 if ("OPTIONS".equalsIgnoreCase(exchange.getRequestMethod())) {
24                     exchange.sendResponseHeaders(204, -1);
25                     exchange.close();
26                     return;
27                 }
28
29                 System.out.println("Handler called! Method: " +
30 exchange.getRequestMethod());
31
32                 try {
33                     // set up Json
34                     String body = new
35 String(exchange.getRequestBody().readAllBytes());
36                     System.out.println("Raw body received: " + body);
37
38                     Gson gson = new Gson();
39                     GameMessage fromClient = gson.fromJson(body, GameMessage.class);
40                     System.out.println("JSON parsed: " + fromClient.boardState + " " +
41 fromClient.move);
42
43                     // set up computer
44                     Computer computer = new Computer(Constants.COMPUTER_MARKER,
45 "COMPUTER");
46
47                     // Create board
48                     Board board = new Board2D(System.out);
49
50                     // Set up message
51                     GameMessage message = new GameMessage();
```

```
47 // read input
48 String boardString = fromClient.boardState;
49 System.out.println("Board received: " + boardString);
50 //Receive client string
51 board.loadString(boardString);
52
53 //Take player move, change from int → position
54 Integer playerMove = Integer.parseInt(fromClient.move);
55 Position playerPos = board.getCellPosition(playerMove);
56 System.out.println("Move received: " + playerMove);
57
58 if(board.isCellEmpty(playerPos)){
59     board.placeMarker(playerPos, Constants.HUMAN_MARKER);
60 } else {
61     //String response = "Cell is occupied!\n" + boardString;
62     String status = "Cell is occupied!";
63     message.status = status;
64     message.boardState = boardString;
65     String json = gson.toJson(message);
66     exchange.sendResponseHeaders(200, json.length());
67     OutputStream out = exchange.getResponseBody();
68     out.write(json.getBytes());
69     out.close();
70     return;
71 }
72
73 //Check winners, continue if don't
74 String status = getGameStatus(board);
75
76 if(status.equals("Computer turns")){
77     board.placeMarker(computer.makeMove(board),
78 Constants.COMPUTER_MARKER);
79     status = getGameStatus(board);
80 }
81
82 // write board
83 String boardState = board.networkString();
84
85 // String response = status + "\n" + board.networkString();
86
87 message.status = status;
88 message.boardState = boardState;
89 String json = gson.toJson(message);
90 exchange.sendResponseHeaders(200, json.length());
91 OutputStream out = exchange.getResponseBody();
92 out.write(json.getBytes());
93 out.close();
94 } catch (Throwable t){
95     System.out.println("ERROR: " + t.getMessage());
96     t.printStackTrace();
97 }
98 });
99 server.start();
```

```
100     System.out.println("Server started on port 9040");
101     }
102
103     // This is me again, get tired of writing long if else :) so I create a
    helper
104     private static String getGameStatus(Board board){
105         char winner = board.checkWinner();
106         if (winner == Constants.HUMAN_MARKER) return "You wins!";
107         if (winner == Constants.COMPUTER_MARKER) return "Computer wins!";
108         if (board.isBoardFull()) return "Draw!";
109         return "Computer turns";
110
111
112     }
113 }
```